CSR Multimodal AI Agent: Clinical Query Report

This report showcases the capabilities of the **CSR Multimodal AI Agent** in extracting, retrieving, and interpreting clinical study data (**Text**, **Tables**, and **Figures**) using Gemini 2.5 Pro.

Query 1: Query 3 - Statistical Analysis Plan & Dose Optimisation

Clinical Question:

Summarise the statistical methodology used in the BNT162b2 efficacy analysis: the primary endpoint definition, the vaccine efficacy calculation formula, multiplicity adjustment strategy, and the criteria used to select the 30 µg dose as the optimal dose over lower doses (1, 3, 10, 20 µg). Include the multiplicity schema figure and any dose-selection tables that show immunogenicity data across dose levels from the Investigator's Brochure and the Statistical Analysis Plan.

AI Analysis:

Based on the clinical study documents provided for the Pfizer-BioNTech BNT162b2 vaccine, here is a summary of the statistical methodology used in the efficacy analysis and the rationale for the 30 µg dose selection.

Summary

The statistical plan for the BNT162b2 clinical trial defined the primary efficacy endpoint as the prevention of confirmed COVID-19 with onset at least 7 days after the second dose. Vaccine Efficacy (VE) was determined by comparing the incidence rate of COVID-19 cases in the vaccine group against the placebo group. The trial employed a detailed multiplicity adjustment strategy to control for error rates across multiple statistical tests, though the provided schema specifically details the strategy for immunogenicity endpoints rather than efficacy. The 30 µg dose of BNT162b2 was selected for the pivotal Phase 2/3 study based on its acceptable tolerability and safety profile observed in Phase 1, where reactogenicity was noted to increase with dose level.

Key Findings and Supporting Evidence

1. Primary Efficacy Endpoint

The primary efficacy endpoint of the study was the prevention of COVID-19 in participants without evidence of prior SARS-CoV-2 infection.

- **Definition**: The primary endpoint was the "First COVID-19 Occurrence From 7 Days After Dose 2" in the evaluable efficacy population (Protocol.pdf, p. 56). This means the analysis counted confirmed cases of COVID-19 that occurred at least one week after a participant had received their second dose of the vaccine or placebo.

2. Vaccine Efficacy (VE) Calculation

Vaccine efficacy is a measure of the proportionate reduction in the incidence of disease in vaccinated individuals compared to unvaccinated individuals.

- **Methodology**: While the specific formula is not provided in the excerpts, the analysis is based on comparing the

number of COVID-19 cases in the BNT162b2 group to the number of cases in the placebo group among the total participants in each arm. The primary analysis table is titled "Vaccine Efficacy - First COVID-19 Occurrence From 7 Days After Dose 2" (Protocol.pdf, p. 56), indicating a comparison of case counts between the randomized groups. VE is conventionally calculated as VE = (1 - Relative Risk) x 100%, where the relative risk is the incidence rate in the vaccine group divided by the incidence rate in the placebo group.

3. Multiplicity Adjustment Strategy

To avoid an increased risk of false-positive results when conducting multiple statistical tests, a multiplicity adjustment strategy was implemented. The provided documentation includes a detailed schema for managing multiplicity for the study's *immunogenicity* endpoints.

- **Multiplicity Schema**: The figure below from the Statistical Analysis Plan (SAP) outlines the strategy for controlling the overall Type I error rate (alpha) at 0.025 for immunogenicity analyses.

> Statistical Analysis Methods (Source: CTP, Page 103)

Endpoint	Statistical Analysis Methods
Secondary immunogenicity	Geometric mean titers/concentrations (GMTs/GMCs) of SARS-CoV-2 neutralizing titers, S1-binding IgG level, and RBD-binding IgG level For SARS-CoV-2 neutralizing titers, S1-binding IgG levels, and RBD-binding IgG levels, GMTs/GMCs and 2-sided 95% CIs will be provided for each investigational product within each group before vaccination and at each of the following time points: - Phase 1: 7 and 21 days after Dose 1; 7 and 14 days and 1, 6, 12 and 24 months after Dose 2 Geometric means will be calculated as the mean of the assay results after making the logarithm transformation and then exponentiating the mean to express results on the original scale. Two-sided 95% CIs will be obtained by taking natural log transforms of concentrations/titers, calculating the 95% CI with reference to the t-distribution, and then exponentiating the confidence limits. GMFRs of SARS-CoV-2 neutralizing titers, S1-binding IgG level, and RBD-binding IgG level For SARS-CoV-2 neutralizing titers, S1-binding IgG levels, and RBD-binding IgG levels, the GMFRs and 2-sided 95% CIs will be provided for each investigational product within each group at each of the following time points: - Phase 1: 7 and 21 days after Dose 1; 7 and 14 days and 1, 6, 12, and 24 months after Dose 2 GMFRs will be limited to participants with nonmissing values prior to the first dose and at the postvaccination time point. The GMFR will be calculated as the mean of the difference of logarithmically transformed assay results (later time point - earlier time point) and exponentiating the mean. The associated 2-sided CIs will be obtained by calculating CIs using Student's t-distribution for the mean difference of the logarithmically transformed assay results and exponentiating the confidence limits. Percentage of participants with 4-fold rise in SARS-CoV-2 neutralizing titers, S1-binding IgG level, and RBD-binding IgG level For SARS-CoV-2 neutralizing titers, S1-binding IgG levels, and RBD-binding IgG levels, percentages (and 2-sided 95% CIs) of

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Endpoint	Statistical Analysis Methods
	participants with 4-fold rise will be provided for each investigational product within each group at each of the following time points: - Phase 1: 7 and 21 days after Dose 1; 7 and 14 days and 1, 6, 12, and 24 months after Dose 2 The Clopper-Pearson method will be used to calculate the CIs. GMR of SARS-CoV-2 neutralizing titer to S1-binding IgG level and to RBD-binding IgG level For SARS-CoV-2 neutralizing titers, S1-binding IgG levels, and RBD-binding IgG levels, the GMRs and 2-sided 95% CIs will be provided for each investigational product within each group at each of the following time points: - Phase 1: 7 and 21 days after Dose 1; 7 and 14 days and 1, 6, 12, and 24 months after Dose 2 GMRs will be limited to participants with nonmissing values for both SARS-CoV-2 neutralizing titers and S1-binding IgG level/RBD- binding IgG level at each time point. The GMR will be calculated as the mean of the difference of logarithmically transformed assay results (eg, SARS-CoV-2 neutralizing titers minus S1-binding IgG level for each participant) and exponentiating the mean. Two-sided CIs will be obtained by calculating CIs using Student's t-distribution for the mean difference of the logarithmically transformed assay results and exponentiating the confidence limits. For all the immunogenicity endpoints, the analysis will be based on the Dose 1 and Dose 2 evaluable immunogenicity populations. An additional analysis will be performed based on the all-available immunogenicity populations if there is a large enough difference in sample size between the all-available immunogenicity populations and the evaluable immunogenicity populations. Participants will be summarized according to the vaccine group to which they were randomized. Missing serology data will not be imputed.
Secondary immunogenicity (noninferiority in the 12- to 15-year age group compared to the	GMR of SARS-CoV-2 neutralizing titers in participants 12 to 15 years of age to those 16 to 25 years of age For participants with no serological or virological evidence (up to 1 month after receipt of the second dose) of past SARS-CoV-2 infection, the GMR of SARS-CoV-2 neutralizing titers in participants 12 to 15 years of age to those in participants 16 to 25 years of age and
16- to 25-year age group)	2-sided 95% CIs will be provided at 1 month after Dose 2 for noninferiority assessment. The GMR and its 2-sided 95% CI will be derived by calculating differences in means and CIs on the natural log scale of the titers based on the Student's t-distribution and then exponentiating the results. The difference in means on the natural log scale will be 12 to 15 years minus 16 to 25 years. Noninferiority will be declared if the lower bound of the 2-sided 95% CI for the GMR is greater than 0.67. This analysis will be based on Dose 2 evaluable immunogenicity populations. An additional analysis may be performed based on the Dose 2 all-available immunogenicity population if needed. Participants will be summarized according to the vaccine group to which they were randomized. Missing serology data will not be imputed.

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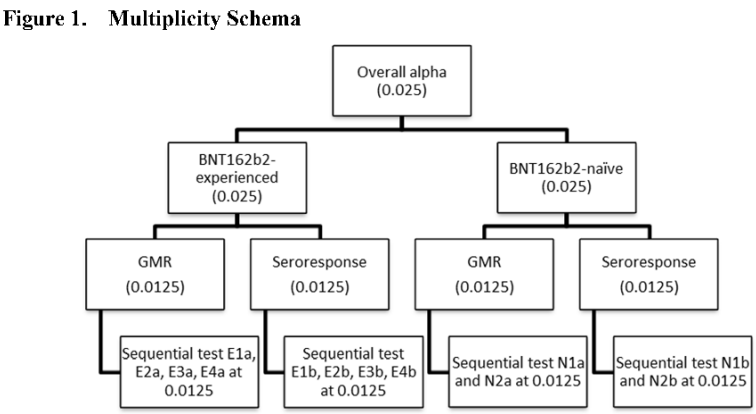
Endpoint	Statistical Analysis Methods
Exploratory immunogenicity	Geometric mean titers/concentrations (GMTs/GMCs) of SARS-CoV-2 neutralizing titers, S1-binding IgG level, and RBD-binding IgG level For SARS-CoV-2 neutralizing titers, S1-binding IgG levels, and RBD-binding IgG levels, GMTs/GMCs and 2-sided 95% CIs will be provided for each investigational product within each group before vaccination and at each of the following time points in Phase 2/3: - 1, 6, 12, and 24 months after completion of vaccination in participants with and without serological or virological evidence of SARS-CoV-2 infection before vaccination Geometric means will be calculated as the mean of the assay results after making the logarithm transformation and then exponentiating the mean to express results on the original scale. Two-sided 95% CIs will be obtained by taking natural log transforms of concentrations/titers, calculating the 95% CI with reference to the t-distribution, and then exponentiating the confidence limits. GMFRs of SARS-CoV-2 neutralizing titers, S1-binding IgG level, and RBD-binding IgG level For SARS-CoV-2 neutralizing titers, S1-binding IgG levels, and RBD-binding IgG levels, the GMFRs and 2-sided 95% CIs will be provided for each investigational product within each group at each of the following time points in Phase 2/3:
	1, 6, 12, and 24 months after completion of vaccination in participants with and without serological or virological evidence of SARS-CoV-2 infection before vaccination GMFRs will be limited to participants with nonmissing values prior to the first dose and at the postvaccination time point. The GMFR will be calculated as the mean of the difference of logarithmically transformed assay results (later time point - earlier time point) and exponentiating the mean. The associated 2-sided CIs will be obtained by calculating CIs using Student's t-distribution for the mean difference of the logarithmically transformed assay results and exponentiating the confidence limits. Percentage of participants with antibody levels predefined threshold(s) for SARS-CoV-2 serological parameters For SARS-CoV-2 neutralizing titers, S1-binding IgG levels and/or RBD-binding IgG levels, N-binding antibody, and SARS-CoV-2 detection by NAAT, percentages (and 2-sided 95% CIs) of participants with antibody levels predefined threshold(s) will be provided for each investigational product within each group at baseline and each of the following time points in Phase 2/3: - 1, 6, 12, and 24 months after completion of vaccination in participants with and without serological or virological evidence of SARS-CoV-2 infection before vaccination The Clopper-Pearson method will be used to calculate the CIs. Percentage of participants with the immune response (non-S) to SARS-CoV-2 for N-binding antibody at the time points when data are available The Clopper-Pearson method will be used to calculate the CIs. For all of the immunogenicity endpoints, the analysis will be based on the Dose 1 and Dose 2 evaluable immunogenicity populations. An additional analysis will be performed based on the all-available immunogenicity populations if there is a large enough difference in sample size between the all-available immunogenicity populations and the evaluable immunogenicity populations. Participants will be summarized according to the vaccine group to which they were randomized. Missing serology data will not be imputed.

Endpoint	Statistical Analysis Methods
	RCDCs for immunogenicity results Empirical RCDCs will be provided for SARS-CoV-2 neutralizing titers, S1-binding IgG level, and RBD-binding IgG level after Dose 1 and after Dose 2.
	The efficacy analysis for the second primary objective evaluation will be based on all participants included in the evaluable efficacy population and in the all-available efficacy population. For the primary endpoint analysis, missing efficacy data will not be imputed. A sensitivity analysis will be performed by imputing missing values with the assumption of MAR. A missing efficacy endpoint may be imputed based on predicted probability using the fully conditional specification method. Other imputation methods without the MAR assumption may be explored. The details will be provided in the SAP.
Secondary	First: Ratio of confirmed COVID-19 illness from 14 days after the second dose per 1000 person-years of follow-up in participants without evidence of infection (prior to 14 days after receipt of the second dose) for the active vaccine group to the placebo group Second: Ratio of confirmed COVID-19 illness from 14 days after the second dose per 1000 person-years of follow-up in participants with and without evidence of infection (prior to 14 days after receipt of the second dose) for the active vaccine group to the placebo group Third and fourth: Ratios of confirmed severe COVID-19 illness from 7 days and from 14 days after the second dose per 1000 person-years of follow-up in participants without evidence of infection (prior to 7 days or 14 days after receipt of the second dose) for the active vaccine group to the placebo group Fifth and sixth: Ratios of confirmed severe COVID-19 illness from 7 days and from 14 days after the second dose per 1000 person-years of follow-up in participants with and without evidence of infection (prior to 7 days or 14 days after receipt of the second dose) for the active vaccine group to the placebo group These secondary efficacy objectives will be evaluated sequentially in the order specified above after the primary objectives are met. The analysis will be based on the evaluable efficacy population and the all- available efficacy population. The analysis methodology used for the primary efficacy endpoints will be applied for the analysis of the above secondary efficacy endpoints. The following secondary efficacy endpoints will be evaluated descriptively with 95% CIs. Ratios of confirmed COVID-19 illness (according to the CDC-defined symptoms) from 7 days and from 14 days after the
	second dose per 1000 person-years of follow-up in participants without evidence of infection (prior to 7 days or 14 days after receipt of the second dose) for the active vaccine group to the placebo group Ratios of confirmed COVID-19 illness (according to the CDC-defined symptoms) from 7 days and from 14 days after the second dose per 1000 person-years of follow-up in participants with and without evidence of infection (prior to 7 days or 14 days after receipt of the second dose) for the active vaccine group to the placebo group VE = 100 × (1 - IRR) will be estimated with confirmed COVID-19 illness according to the CDC-defined symptoms from 7 days or from 14 days after the second dose. The 2-sided 95% CI for VE will be derived using the Clopper-Pearson method as described by Agresti.9 Missing efficacy data will not be imputed.

Endpoint	Statistical Analysis Methods
	review plan; (2) Tier 2 events are those that are not Tier 1 but are considered "relatively common"; a MedDRA preferred term is defined as a Tier 2 event if there are at least 1% of participants in at least 1 vaccine group reporting the event; and (3) Tier 3 events are those that are neither Tier 1 nor Tier 2 events. For both Tier 1 and Tier 2 events, 2-sided 95% CIs for the difference between the vaccine and placebo groups in the percentage of participants reporting the events based on the Miettinen and Nurminen method10 will be provided. In addition, for Tier 1 events, the asymptotic p-values will also be presented for the difference between groups in the percentage of participants reporting the events, based on the same test statistic and under the assumption that the test statistic is asymptotically normally distributed. Descriptive summary statistics (counts, percentages, and associated Clopper-Pearson 95% CIs) will be provided for any AE events for each vaccine group. SAEs will be categorized according to MedDRA terms. Counts, percentages, and the associated Clopper-Pearson 95% CIs of SAEs from Dose 1 to 6 months after the last dose will be provided for each vaccine group. The safety analyses are based on the safety population. Participants will be summarized by vaccine group according to the investigational products they actually received. Missing reactogenicity e-diary data will not be imputed; missing AE dates will be handled according to the Pfizer safety rules.
Secondary	Not applicable (N/A)
Exploratory	N/A

Figure 1. Multiplicity Schema

- [SAP] Figure 1.Multiplicity Schema (Page 48)



!Figure 1. Multiplicity Schema from SAP.pdf page 48

(Source: SAP.pdf, p. 48)

- **Interpretation of the Schema**:
  - The overall alpha for the analysis is set at 0.025.
  - The analysis is split into two parallel paths for two different populations: "BNT162b2-experienced" and "BNT162b2-naive" participants.
  - Within each population, the alpha is further split (0.0125 each) between two types of immunogenicity endpoints: Geometric Mean Ratio (GMR) and Seroresponse.
  - A sequential testing procedure is used for hypotheses within each endpoint type (e.g., "Sequential test N1a and N2a at 0.0125"). This means the first hypothesis (N1a) is tested at the 0.0125 level, and only if it is statistically significant is the next hypothesis (N2a) formally tested. This approach preserves the overall alpha for the family of tests.

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It is important to note that this specific schema applies to the immunogenicity endpoints (antibody levels), not the clinical efficacy endpoints (disease prevention).

4. Rationale for 30 µg Dose Selection

The decision to proceed with the 30 µg dose level for the Phase 2/3 efficacy study was based on Phase 1 data, which balanced the vaccine's safety and tolerability against its ability to generate an immune response (immunogenicity).

- **Confirmation of Selection**: A protocol amendment confirms the final choice: "the vaccine candidate selected for Phase 2/3 evaluation is BNT162b2 at a dose of 30 µg" (CTP.pdf, p. 40).

- **Tolerability and Safety Profile**:
  - BNT162b2 was tested at dose levels between 1 µg and 30 µg in younger adults, and "acceptable tolerability was shown after both doses at all dose levels" (IB.pdf, p. 11).
  - Reactogenicity (local and systemic reactions) was found to increase with higher dose levels. For the BNT162b1 candidate (which had a similar profile), analysis of the 10 µg, 20 µg, and 30 µg doses showed that "pain at the injection site was the most frequent prompted local reaction, increasing in frequency and/or severity with increasing dose level" and "reactogenicity increased with increasing dose level" (IB.pdf, p. 69).
  - The 30 µg dose was therefore considered the optimal dose that provided a strong immune response while maintaining an acceptable tolerability profile suitable for a large-scale vaccination program. The provided documents do not contain the comparative immunogenicity data tables across all tested dose levels that would quantitatively demonstrate the superior immune response of the 30 µg dose compared to lower doses.

Sources

- **CTP.pdf**: Protocol C4591001, Page 40
- **IB.pdf**: Investigator's Brochure, Pages 11, 69
- **SAP.pdf**: Statistical Analysis Plan, Page 48
- **Protocol.pdf**: Protocol Document, Page 56

Additional Tables Consulted:

> demographic characteristics. The efficacy, safety, and immunogenicity data in this EUA (Source: Protocol, Page 75)

		COVID-19		**Vaccine**										
VRBPAC	Briefing		Document											
highly	effective		protection	against	COVID-19		in	a broad	population		of individuals		across	
demographic		characteristics.			The efficacy,		safety,	and	immunogenicity		data	in this	EUA	
application		support	a positive		assessment		of risks	and	benefits	for the				COVID-
19 Vaccine		and fulfil	the	data	requirements		outlined	in	the FDA	EUA	guidance.		The	
registrational		Phase	1/2/3	Study,	C4591001,		will	continue	as	long	as possible	per	protocol	to
monitor	participants		and	obtain	additional		data to	support	a BLA	filing	in	the near	future.	An
extensive		post-authorization			plan has	been	prepared	to	monitor	and	evaluate	the	safety	of
BioNTech		COVID-19		Vaccine	in persons		vaccinated		under	an EUA.				

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		COVID-19		**Vaccine**										
REFERENCES														
1	Food	and Drug	Administration			Emergency		Use Authorization			for Vaccines		to Prevent	
	COVID-19.		October	2020.	Available		at: https://www.fda.gov/media/142749/download							
2	Virus	Pathogen	Database		and Analysis		Resource		(ViPR).	Available		at:		
	https://www.viprbrc.org/brc/viprStrainDetails.spgncbiAccession=MN908947&decorator=c													
	orona.	Accessed	11	August	2020.									
3	Zhu N,	Zhang	D,	Wang	W, et	al. A novel	coronavirus		from	patients		with pneumonia		in
	China,	2019.	N Engl	J Med	2020;382:727-33.									
4	Johns	Hopkins	University		Coronavirus		Resource		Center.	Available	at:			
	https://coronavirus.jhu.edu/map.html.						Accessed	17	November		2020.			
5	FDA	Guidance	for	Industry	Expedited		Programs		for Serious	Conditions--Drugs			and	

		COVID-19		**Vaccine**										
	Biologics.	May	2014.	Available		at: https://www.fda.gov/media/86377/download								
6	FDA	Guidance	for	Industry	on	E2E Pharmacovigilance				Planning	(2005).	Available		at:
	https://www.fda.gov/media/71238/download													
7	Food	and Drug	Administration			Development		and	Licensure	of	Vaccines	to	Prevent	COVID-
	19 Guidance		for Industry.		June	2020.	Available	at:						
	https://www.fda.gov/media/139638/download													
8	Centers	for	Disease	Control	and	Prevention		guidance	on	Pandemic	Influenza.		Available	at:
	https://www.cdc.gov/flu/pandemic-resources/index.htm													
9	World	Health	Organization		Coronavirus		Disease		(COVID-1)		Dashboard.	Accessed		16
	November	2020.	Available		at:	https://covid19.who.int/								
10	Sui J,	W Li,	A Murakami,		et al.	Potent	neutralization		of	severe	acute	respiratory	syndrome	
	(SARS)	coronavirus		by	a human	mAb	to S1	protein	that	blocks	receptor	association.		PNAS
	2004;	101:2536-41.												

		COVID-19		**Vaccine**										
11	Jiang,L,	N Wang,		T Zuo	et al.	Potent	Neutralization		of	MERS-CoV		by Human	Neutralizing	
	Monoclonal		Antibodies		to the	Viral Spike	Glycoprotein.			Sci Transl	Med	2014;6:234ra9.		
							Page	75						
VRBPAC	Briefing	Document												
12	Wrapp	D, Wang	N,	Corbett	KS,	et al.	Cryo-EM	structure		of the	2019-nCoV		spike	in the
	prefusion	conformation.		Science		2020;367(6483):1260-63.								
13	Pallesen	J, Wang	N,	Corbett	KS,	et al.	Immunogenicity		and	structures		of a rationally		
	designed	prefusion		MERS-CoV		spike	antigen.	Proc	Natl	Acad Sci	2017;114(35):E7348-57.			
14	Maruggi	G, Zhang		C, Li	J, et al.	mRNA	as a	Transformative		Technology		for	Vaccine	
	Development		to Control	Infectious		Diseases.		Mol	Ther 2019;27(4):757-772.					
15	Sahay	G, Alakhova		DY, Kabanov		AV.	Endocytosis		of nanomedicines.			J Control	Release	
	2010;145(3):182-195.													
16	Lambert,	P.-H.	et al.	Consensus		summary	report	for	CEPI/BC	March	12-13,		2020	meeting:
	Assessment	of	risk	of disease	enhancement			with COVID-19		vaccines.		Vaccine	38,	4783-
	4791;	10.1016/j.vaccine.2020.05.064					(2020).							
17	Vogel	AB, Kanevsky		I, Che	Y,	et al. A	prefusion	SARS-CoV-2			spike	RNA	vaccine	is
	highly	immunogenic		and	prevents	lung	infection	in	non-human		primate.	2020;	published	

		COVID-19		**Vaccine**										
	online	at https://www.biorxiv.org/content/10.1101/2020.09.08.280818v1.												
18	Walsh	EE, Frenck		RW Jr,	Falsey	AR,	et al.	Safety	and Immunogenicity			of Two	RNA-Based	
	Covid-19	Vaccine		Candidates.		N Engl	J Med.	2020;[ePub]:doi:			10.1056/NEJMoa2027906.			
19	Mulligan	MJ,	Lyke	KE, Kitchin		N, et	al. Phase	I/II	study	of COVID-19		RNA	vaccine	
	BNT162b1	in	adults.	Nature	2020;586(7830):589-93.				[ePub]	doi:	10.1038/s41586-020-			
	2639-4.													
20	Baum,	A. et	al. Antibody		cocktail	to SARS-CoV-2			spike	protein	prevents	rapid	mutational	
	escape	seen with	individual		antibodies.		Science	369,	1014-1018		(2020).			
21	Zhang,	L. et	al. The	D614G	mutation		in the	SARS-CoV-2		spike	protein	reduces	S1	
	shedding	and	increases	infectivity.		bioRxiv		Prepr.	Serv.	Biol. (2020).		doi:bioRxiv:		
	10.1101/2020.06.12.148726													
22	Kranz	LM, Diken	M,	Haas	H, et	al. Systemic		RNA	delivery	to dendritic		cells	exploits	
	antiviral	defence	for	cancer	immunotherapy.			Nature	2016;534,	396-401.				
23	Kamphuis	E,	Junt T,	Waibler	Z,	et al.	Type	I interferons		directly	regulate	lymphocyte		
	recirculation		and cause	transient		blood	lymphopenia.		Blood	2006;108(10):3253-61.				

		COVID-19		**Vaccine**										
24	Sahin	U, Muik	A, Derhovanessian			E,	et al. COVID-19		vaccine	BNT162b1		elicits	human	
	antibody	and	Th1 T	cell responses.		Nature	2020;586:594-599.							
25	Institute	for Health		Metrics	and	Evaluation.		Estimates	from	Dec	2020-Feb	2021.	Available	
	at: https://covid19.healthdata.org/united-states-of-america view=total-deaths&tab=trend													
							Page	76						
VRBPAC	Briefing	Document												
26	Biggerstaff	M.	Oct 2020	ACIP	meeting.		Available	at:						
	https://www.cdc.gov/vaccines/acip/meetings/slides-2020-10.html.													
27	Biggerstaff	M,	Reed	C, Swerdlow		D, et	al. Estimating		the potential	effects	of a	vaccine		
	program	against	an emerging		influenza	pandemic--United			States.	Clinical	Infectious			
	Diseases	2015;60(S1):S20-9.												
28	Anderson	R, Vegvari		C, Truscott		J, Collyer	BS.	Challenges	in	creating	herd immunity	to		

		COVID-19		**Vaccine**										
	SARS-CoV-2		infection	by	mass vaccination.		Lancet	2020;[ePub]:doi.org/10.1016/				S0140-		
	6736(20)32318-7.													
							Page 77							

> Table 10. Vaccine Efficacy - First COVID-19 Occurrence From 7 Days After Dose 2 - (Source: Protocol, Page 56)

Table 10. Vaccine Efficacy - First COVID-19 Occurrence From 7 Days After Dose 2 - Subjects With or Without Evidence of Infection Prior to 7 Days After Dose 2 - Evaluable Efficacy (7 Days) Population			
	Vaccine Group (as Randomized)		
	BNT162b2 (30 ug) (Na=19965)	Placebo (Na=20172)	
First COVID-19 occurrence from 7 9 2.332 (18559) 169 2.345 (18708) 94.6 (89.9, >0.9999 days after Dose 2 97.3)			

Table 10. Vaccine Efficacy - First COVID-19 Occurrence From 7 Days After Dose 2 - Subjects With or Without Evidence of Infection Prior to 7 Days After Dose 2 - Evaluable Efficacy (7 Days) Population			
Abbreviations: VE = vaccine efficacy. a. N = number of subjects in the specified group. b. n1 = Number of subjects meeting the endpoint definition. c. Total surveillance time in 1000 person-years for the given endpoint across all subjects within each group at risk for the endpoint. Time period for COVID-19 case accrual is from 7 days after Dose 2 to the end of the surveillance period. d. n2 = Number of subjects at risk for the endpoint. e. Credible interval for VE was calculated using a beta-binomial model with prior beta (0.700102, 1) adjusted for surveillance time. Refer to the statistical analysis plan, Appendix 2, for more details. f. Posterior probability (Pr) was calculated using a beta-binomial model with prior beta (0.700102, 1) adjusted for surveillance time. Refer to the statistical analysis plan, Appendix 2, for more details. PFIZER CONFIDENTIAL SDTM Creation: 17NOV2020 (09:48) Source Data: adc19ef Table Generation: 17NOV2020 (16:46) (Cutoff Date: 14NOV2020, Snapshot Date: 16NOV2020) Output File: ./nda2unblinded/C4591001EfficacyFA164/adc19efvecov7pd2eval			

Additional Figures Consulted:

- [IB] Figure 22: IFNy ELISpot data for 5 subjects dosed with 10 µg BNT162b2 (BNT162-01) (Page 57)

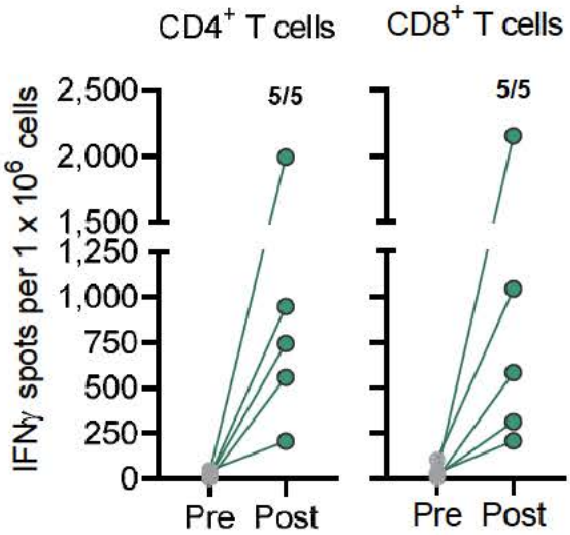

Figure 22: IFNγ ELISpot data for 5 subjects dosed with 10 µg BNT162b2 (BNT162-01)

- [Reports] Figure 9. Cumulative Incidence Curves for the First COVID-19 Occurrence After Dose 1 - Dose 1 All-Available Efficacy Po

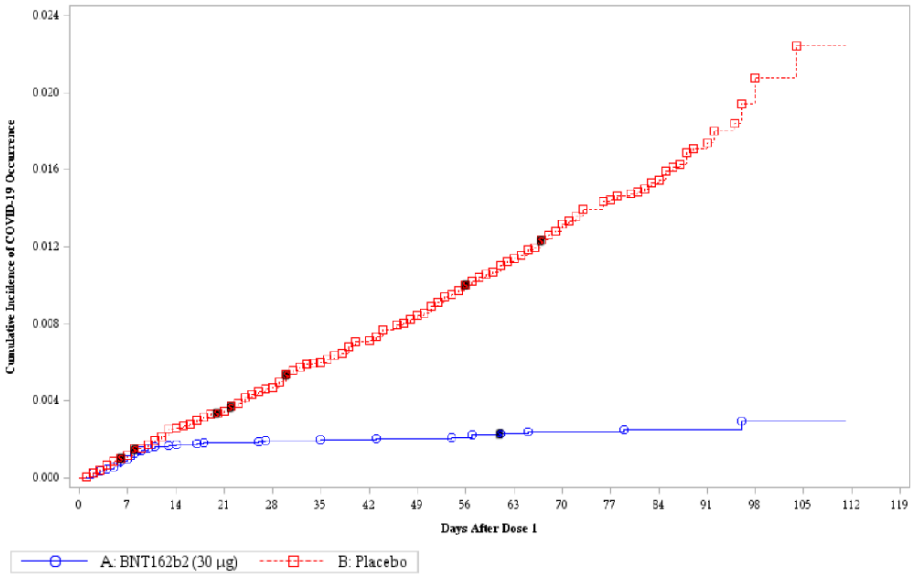

Figure 9. Cumulative Incidence Curves for the First COVID-19 Occurrence After Dose 1 – Dose 1 All-Available Efficacy Population

Citations:

CTP.pdf (pg 40), CTP.pdf (pg 103), IB.pdf (pg 11), IB.pdf (pg 57), IB.pdf (pg 69), Protocol.pdf (pg 56), Protocol.pdf (pg 75), Reports.pdf (pg 82), SAP.pdf (pg 48), SAP.pdf (pg 67)